BOILEAU GUILLAUME

Ingénieur R&D Physique, Signal & IA | Détection faible SNR, Python, Fusion de données, Validation terrain

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EXPERIENCE

Software Engineer – System Integration, Data Flows & Operational Validation

VEV Platform Services France

2025 – 2026 · Valbonne, France

Context: Industrial software environment for an EV charging CPMS platform connected to operational charging stations, field equipment and hardware/software interfaces.
Objective: Support reliable operation of connected infrastructure through software integration, data-flow checks, operational monitoring, anomaly analysis and validation of technical behaviours.

- **Operational data flows:** Implemented and monitored FTP-based exchanges between software services and external systems.
- **Anomaly analysis:** Investigated interrupted exchanges, inconsistent flows and defects affecting service behaviour.
- **Validation:** Performed checks in production-like environments to assess continuity, expected behaviour and data consistency.
- **Software support:** Maintained TypeScript, Angular and Node.js components and documented technical findings.

Senior R&D Engineer – Signal Processing, Modelling & Validation Workflows

CNES, Laboratoire Lagrange, Observatoire de la Côte d'Azur

2023 – 2025 · Nice, France

Context: R&D activities for the LISA ESA/NASA space mission, involving physical modelling, weak-signal analysis, simulation, software chains, data-processing workflows and validation campaigns.
Objective: Develop and validate scientific analysis workflows able to compare complex models, process large datasets and assess weak physical signatures under strong noise and consistency constraints.

- **Weak-signal analysis:** Worked on low signal-to-noise physical backgrounds and complex superposed signals.
- **Signal processing:** Developed methods for spectral interpretation, model comparison and performance assessment.
- **Python scientific platform:** Developed CATpyLISA for large-scale model integration, comparison and validation. GitLab: CATpyLISA
- **Validation campaigns:** Prepared and analysed comparison campaigns on simulated outputs and heterogeneous models.
- **Reproducible workflows:** Used Python, Linux, Git/GitLab and Jupyter-style environments for traceable analyses.
- **Collaboration:** Worked with international contributors across software, systems, modelling and instrumentation.

Data Scientist – Noise Analysis, Statistical Modelling & Performance Validation

Universiteit Antwerpen, Belgium

2021 – 2023 · Antwerp, Belgium

Context: R&D work for Einstein Telescope and next-generation gravitational-wave detector studies, combining environmental noise analysis, detector-configuration studies, statistical modelling and weak-signal detection.
Objective: Analyse the impact of environmental noise on detector sensitivity, develop statistical workflows for different detector configurations and support technical recommendations on noise budgets, geometry and experimental validation.

- **Geophysical data analysis:** Used seismometer measurements to analyse propagation effects and spatial correlations.
- **Noise studies:** Investigated seismic and Newtonian noise impacts on detector sensitivity.
- **Experimental preparation:** Contributed to protocols, sensor selection and deployment preparation for low-noise campaigns.
- **Scientific Python workflows:** Developed reproducible pipelines for signal analysis, statistical modelling and validation.

Junior R&D Engineer – Simulation, Signal Diagnostics & Scientific Computing

ARTEMIS, Observatoire de la Côte d'Azur

2018 – 2021 · Nice, France

Context: Software and analysis activities for gravitational-wave mission preparation, involving simulation, signal decomposition, diagnostic analysis and validation of complex physical environments.
Objective: Develop robust analysis workflows for weak-signal detection, simulation, uncertainty studies and interpretation of complex datasets.

PROFILE FOR INTHEAIR

R&D engineer and PhD physicist with experience in signal processing, noise analysis, scientific Python and validation of complex physical models. Background developed across space systems, precision instrumentation, connected infrastructure and scientific software.
Transferable strengths for this role: physical modelling, low-SNR analysis, reproducible data workflows, experimental reasoning and validation of complex measurements.

- Strong background in physics, modelling and interpretation of complex signals.
- Experience in low-SNR analysis, noise diagnostics and performance assessment.
- Python development for reproducible data-processing and validation workflows.
- Used to working across data, software, physical systems and field constraints.
- Able to ramp up quickly on GPR, geophysics and drone-based data workflows.

LANGUAGES

English	●●●●○
Spanish	●●○○○

EDUCATION

PhD in Physics

Université Côte d'Azur

2018 – 2021 · Nice, France

Selective Graduate Program in Fundamental Physics (Magistère)

Université Paris-Saclay

2016 – 2018 · Orsay, France

MSc in Astronomy and Astrophysics

Observatoire de Paris / Université Paris-Saclay

2017 – 2018 · Paris, France

BSc in Fundamental Physics

Université Paris-Saclay

2015 – 2016 · Orsay, France

PROFESSIONAL TRAINING

Machine Learning in Python with scikit-learn

FUN MOOC – Inria

2026 · France Université Numérique

Predictive modelling, supervised learning, model evaluation, cross-validation, metrics, pipelines and practical machine learning workflows with Python and scikit-learn.

Supervised Machine Learning (Regression & Classification)

DeepLearning.AI – Andrew Ng

2020 · Coursera

- **Signal decomposition:** Developed methods to separate overlapping low-SNR signals.
- **Simulation:** Built approaches for complex signal environments and uncertainty studies.
- **Diagnostics:** Investigated noise sources through cross-sensor correlation analyses.
- **Scientific Python:** Developed Python-based analysis, modelling and reproducible workflows.

Recherche reproductible : principes méthodologi pour une science transparente

FUN MOOC – Inria

2025 — France Université Numérique

Continuous Professional Training – Software, Data, AI and Systems

OpenClassrooms

2024 – 2026

Python, SQL, Machine Learning, AI, Linux, JavaScript, TypeScript, Angular, Node.js, Django, REST APIs, C/C++, web development, algorithms and software engineering fundamentals.

PROJECTS

LISA / CATpyLISA – Weak-Signal Analysis, Model Validation & Scientific Python

ESA/NASA Space Mission – Large-scale International Collaboration

2023 – 2025 — Nice, France

Python workflows for weak-signal analysis, model comparison and validation of complex physical data.

Noise Diagnostics & Performance Analysis for Precision Instrumentation

Signal Processing / Statistical Modelling / Environmental Effects

2021 – 2023 — Antwerp, Belgium

Statistical and signal-analysis workflows to identify environmental and instrumental noise sources.

VEV – Connected Infrastructure, Operational Data Flows & System Validation

Industrial Software / Field Equipment Interfaces

2025 – 2026 — Valbonne, France

Software integration, operational monitoring and validation for a platform connected to field infrastructure.

STRENGTHS

Signal processing & weak-signal detection

Signal Processing · Weak-Signal Detection · Low SNR Analysis · Noise Reduction · Spectral Analysis · Filtering · Correlation Analysis · Signal Decomposition · Anomaly Detection · Performance Analysis · Uncertainty Analysis · Model Comparison

Physics, radar & geophysics ramp-up

Physics · Electromagnetic Modelling Awareness · Propagation Effects · Radar / GPR Ramp-Up · Radargram Interpretation Ramp-Up · Near-Surface Geophysics Ramp-Up · Subsurface Mapping Ramp-Up · Sensor Calibration Logic · Experimental Protocols · Field Validation · Physical Modelling

Python, data science & AI

Python · NumPy · SciPy · pandas · scikit-learn · Matplotlib · Jupyter · Machine Learning · Data Cleaning · Feature Extraction · Classification · Regression · Scientific Computing · Reproducible Workflows

Geospatial & multi-sensor data

Data Fusion · Multi-Source Data · Geospatial Data Ramp-Up · GIS Ramp-Up · QGIS Ramp-Up · LiDAR Awareness · Photogrammetry Awareness · GNSS RTK Awareness · Digital Twins Awareness · Mapping Workflows · Coordinate Systems Ramp-Up · Data Georeferencing Ramp-Up

Validation, software & industrialisation

R&D Validation · Validation Strategy · Test Protocols · Field-Test Logic · Technical Reporting · Traceability · Quality Follow-Up · Linux · Git / GitLab · GitHub · Docker · CI/CD · Bash · SQL · C · C++

Collaboration & R&D delivery

Technical English · International Environment · Stakeholder Coordination · Cross-Functional Collaboration · Agile · Kanban · Backlog Follow-Up · Jira · Confluence · Zenhub · Research-to-Product · Deeptech · Industrial Software · Field Constraints · Autonomy